

```
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```

```
-- CREATE DATABASE
```

```
-- =====
```

```
CREATE DATABASE join_demo;
```

```
USE join_demo;
```

```
-- =====
```

```
-- CREATE DEPARTMENTS TABLE
```

```
-- =====
```

```
CREATE TABLE departments (
```

```
    dept_id INT PRIMARY KEY,
```

```
    dept_name VARCHAR(100)
```

```
);
```

```
-- =====
```

```
-- CREATE EMPLOYEES TABLE
```

```
-- =====
```

```
CREATE TABLE employees (
```

```
    emp_id INT PRIMARY KEY,
```

```
    emp_name VARCHAR(100),
```

```
    dept_id INT,
```

```
    salary DECIMAL(10,2),
```

```
    FOREIGN KEY (dept_id) REFERENCES departments(dept_id)
```

```
);
```

-- =====

-- INSERT INTO DEPARTMENTS

-- =====

INSERT INTO departments VALUES

(1, 'HR'),

(2, 'IT'),

(3, 'Finance'),

(4, 'Marketing');

1 • SELECT * FROM join_demo.departments;

Result Grid		Filter Rows:	Edit:
dept_id	dept_name		
1	HR		
2	IT		
3	Finance		
4	Marketing		
*	NULL		

-- =====

-- INSERT INTO EMPLOYEES

-- =====

INSERT INTO employees VALUES

(101, 'Aung Aung', 1, 800000),

(102, 'Su Su', 2, 900000),

(103, 'Kyaw Kyaw', 2, 850000),

(104, 'Hla Hla', null, 700000), -- No department

(105, 'Mg Mg', 3, 750000); -- Invalid dept (no match)

```
1 • SELECT * FROM join_demo.employees;
```

	emp_id	emp_name	dept_id	salary
▶	101	Aung Aung	1	800000.00
	102	Su Su	2	900000.00
	103	Kyaw Kyaw	2	850000.00
	104	Hla Hla	NULL	700000.00
	105	Mq Mq	3	750000.00
*	NULL	NULL	NULL	NULL

-- =====

-- INNER JOIN

-- Only employees with valid departments

-- =====

```
48 • SELECT e.emp_id, e.emp_name, d.dept_name
49 FROM employees e
50 INNER JOIN departments d
51 ON e.dept_id = d.dept_id;
```

	emp_id	emp_name	dept_name
▶	101	Aung Aung	HR
	102	Su Su	IT
	103	Kyaw Kyaw	IT
	105	Mq Mq	Finance

-- =====

-- LEFT JOIN

-- All employees, even without department

-- =====

```
57 • SELECT e.emp_id, e.emp_name, d.dept_name
58 FROM employees e
59 LEFT JOIN departments d
60 ON e.dept_id = d.dept_id;
```

Result Grid			
Filter Rows:			
Export: Wrap Cell Content: ☐			
	emp_id	emp_name	dept_name
▶	101	Aung Aung	HR
	102	Su Su	IT
	103	Kyaw Kyaw	IT
	104	Hla Hla	NULL
	105	Mq Mq	Finance

-- =====

-- RIGHT JOIN

-- All departments, even without employees

-- =====

```
66 • SELECT e.emp_id, e.emp_name, d.dept_name
67 FROM employees e
68 RIGHT JOIN departments d
69 ON e.dept_id = d.dept_id;
```

Result Grid			
Filter Rows:			
Export: Wrap Cell Content: ☐			
	emp_id	emp_name	dept_name
▶	101	Aung Aung	HR
	102	Su Su	IT
	103	Kyaw Kyaw	IT
	105	Mq Mq	Finance
	NULL	NULL	Marketing

-- =====

-- FULL OUTER JOIN (using UNION)

-- All employees + all departments

-- =====

```
75 • SELECT e.emp_id, e.emp_name, d.dept_name
76 FROM employees e
77 LEFT JOIN departments d
78 ON e.dept_id = d.dept_id
79
80 UNION
81
82 SELECT e.emp_id, e.emp_name, d.dept_name
83 FROM employees e
84 RIGHT JOIN departments d
85 ON e.dept_id = d.dept_id;
86
```

Result Grid				Filter Rows:	Export:	Wrap Cell Content:
	emp_id	emp_name	dept_name			
▶	101	Aung Aung	HR			
	102	Su Su	IT			
	103	Kyaw Kyaw	IT			
	104	Hla Hla	NULL			
	105	Mq Mq	Finance			
	NULL	NULL	Marketing			

-- =====

-- CROSS JOIN

-- Every employee with every department

-- =====

```
91 • SELECT e.emp_name, d.dept_name
92 FROM employees e
93 CROSS JOIN departments d;
```

Result Grid			Filter Rows:	Export:
	emp_name	dept_name		
▶	Aung Aung	Marketing		
	Aung Aung	Finance		
	Aung Aung	IT		
	Aung Aung	HR		
	Su Su	Marketing		
	Su Su	Finance		
	Su Su	IT		
	Su Su	HR		
	Kyaw Kyaw	Marketing		
	Kyaw Kyaw	Finance		
	Kyaw Kyaw	IT		
	Kyaw Kyaw	HR		
	Hla Hla	Marketing		
	Hla Hla	Finance		
	Hla Hla	IT		
	Hla Hla	HR		
	Mq Mq	Marketing		
	Mq Mq	Finance		
	Mq Mq	IT		
	Mq Mq	HR		

-- =====

-- SELF JOIN (Manager Example)

-- =====

-- Add manager_id column first

ALTER TABLE employees ADD manager_id INT;

-- Update sample manager data

UPDATE employees SET manager_id = 101 WHERE emp_id = 102;

UPDATE employees SET manager_id = 101 WHERE emp_id = 103;

-- Query

```
106 • SELECT e.emp_name AS Employee,  
107         m.emp_name AS Manager  
108 FROM employees e  
109 LEFT JOIN employees m  
110 ON e.manager_id = m.emp_id;
```

Result Grid			Filter Rows:	Export:
	Employee	Manager		
▶	Aung Aung	NULL		
	Su Su	Aung Aung		
	Kyaw Kyaw	Aung Aung		
	Hla Hla	NULL		
	Mq Mq	NULL		

-- =====

-- NATURAL JOIN

-- (dept_id is common column)

-- =====

```
116 • SELECT *
117 FROM employees
118 NATURAL JOIN departments;
```

Result Grid						
		Filter Rows:		Export:	Wrap Cell Content:	
	dept_id	emp_id	emp_name	salary	manager_id	dept_name
▶	1	101	Aung Aung	800000.00	NULL	HR
	2	102	Su Su	900000.00	101	IT
	2	103	Kyaw Kyaw	850000.00	101	IT
	3	105	Mg Mg	750000.00	NULL	Finance

-- =====

-- JOIN with condition

-- =====

```
123 • SELECT e.emp_name, d.dept_name, e.salary
124 FROM employees e
125 INNER JOIN departments d
126 ON e.dept_id = d.dept_id
127 WHERE e.salary > 800000;
128
```

Result Grid			
		Filter Rows:	
		Export:	Wrap Cell Content:
	emp_name	dept_name	salary
▶	Su Su	IT	900000.00
	Kyaw Kyaw	IT	850000.00